26900126 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECND303 - FOOD PRESERVATION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which of the following is considered a traditional (non-emerging) method of food preservation? [An] / [CO1]

   a). High Pressure Processing

   b). Salting

   c). Ohmic Heating

   d). Plasma Light Technology

2. Cellar storage is mainly based on controlling: [U] / [CO3]

   a). Temperature and environmental conditions

   b). Temperature and environmental conditions

   c). Radiation

d). Salt concentration

3. Salt preserves food mainly by: [K] / [CO4]

a). Reducing water activity through osmosis

b). Increasing moisture

c). Increasing microbial growth

d). Promoting microbial growth

4. Fermentation is a process involving the action of: [K] / [CO5]

a). Microorganisms

b). Microorganisms

c). Natural pigment

d). Antioxidant nutrient

5. Which of the following is an antioxidant? [K] / [CO5]

a). Ascorbic acid

b). Sodium chloride

c). Increase cooking time

d). Reduce nutritional value

6. The rate of heating in Ohmic heating depends primarily on: [An] / [CO2]

a). The electrical conductivity of the food

b). The colour of the food

c). The pH of the food

d). The packaging material

7. The primary purpose of canning is to: [U] / [CO3]

a). Increase water activity

b). Destroy microorganisms and prevent recontamination

c). Increase water activity

d). Promote fermentation

8. Which of the following is a natural method of drying food? [K] / [CO4]

a). Solar drying

b). Spray drying

c). Drum drying d). Canning followed by freezing

9. Pectin is particularly important in the preparation of: [K] / [CO4]

a). Jelly b). Pickle

c). Cured meat d). Marmalade

10. Which product is commonly prepared using spray drying? [K] / [CO4]

a). Milk powder b). Pickle

c). Cured meat d). Cured meat

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. What is lactic acid fermentation? [U] / [CO5]

12. What is meant by "non-thermal" food preservation? Give one example. [U] / [CO2]

13. Mention any two substances commonly used in meat curing. [K] / [CO4]

14. What are chemical preservatives? [K] / [CO5]

15. What is the Radura symbol used for? [U] / [CO2]

16. Differentiate between food irradiation and E-beam irradiation. [An] / [CO2]

17. What is meant by "commercial sterility" in canning? [An] / [CO3]

[2x5 = 10]

Part C

Short Essay

Answer any **6** questions

Each question carries **5** marks

18. Explain the principle of dehydrofreezing and mention its advantages. [U] / [CO4]

19. Explain how fermentation improves the nutritional and safety profile of food. [U] / [CO5]

20. Explain Ohmic Heating and describe its mechanism of internal heat generation. [U] / [CO2]

21. Explain freeze drying (lyophilization) and describe its principle and process. [U] / [CO3]

22. Explain the classification of chemical preservatives into Class I and Class II with examples. [U] / [CO5]

23. Explain pasteurization and describe its importance in food preservation. [U] / [CO3]

24. Discuss the importance of chemical preservatives in maintaining food quality and extending shelf life. [An] / [CO5]

25. Define fermentation and explain its importance in food preservation. [U] / [CO5]

[5x6 = 30]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26. Compare and contrast conventional thermal preservation methods with emerging non-thermal technologies (HPP, PEF, Ultrasonic, Ohmic Heating), highlighting their impact on microbial safety and food quality. [An] / [CO2]

27. Analyse the different methods of preservation by low temperature. [U] / [CO3]

28. Explain the principle and process of foam mat drying and discuss its advantages, limitations and applications in food preservation. [An] / [CO4]

29. Describe mechanical drying in detail, explaining its principle, working, important operating factors, advantages and limitations. [An] / [CO4]

[10x2 = 20]